<u>Forklift Safety Alert</u>

Implement ultrasonic sensor in forklift for pedestrian detection; activate buzzer and stop motor if object within 2m.

INTRODUCTION:

A forklift safety alert system is designed to improve safety in industrial environments by preventing accidents between forklifts and pedestrians. Using an ultrasonic sensor, the system continuously measures the distance between the forklift and nearby objects or workers. If a person or obstacle is detected within a predefined safety range (2 meters), the system immediately alerts the driver using a buzzer and automatically stops the motor to prevent collisions.

OBJECTIVE:

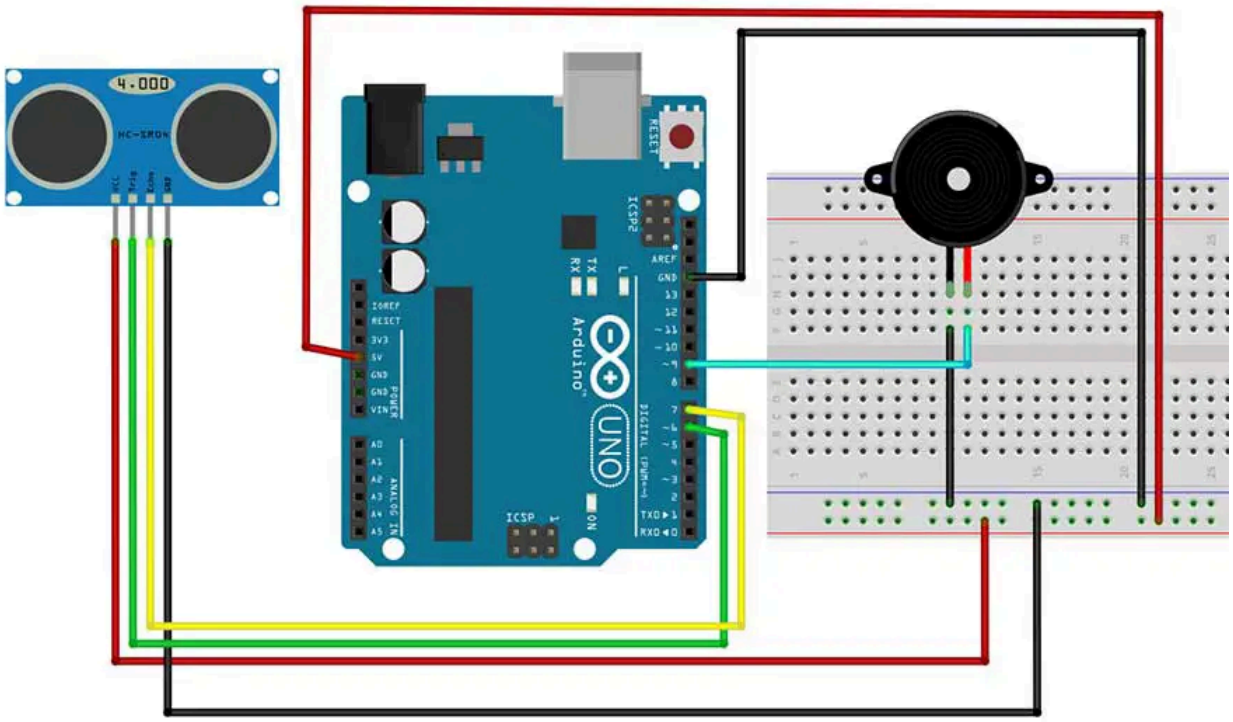
- To detect nearby obstacles or pedestrians using an ultrasonic sensor
- To create a safety boundary of 2 meters around the forklift
- To alert the driver using a buzzer system
- To automatically stop the forklift motor when danger is detected
- To enhance industrial safety and accident prevention
- To simulate real-time safety control using Arduino

TECHNOLOGY USED:

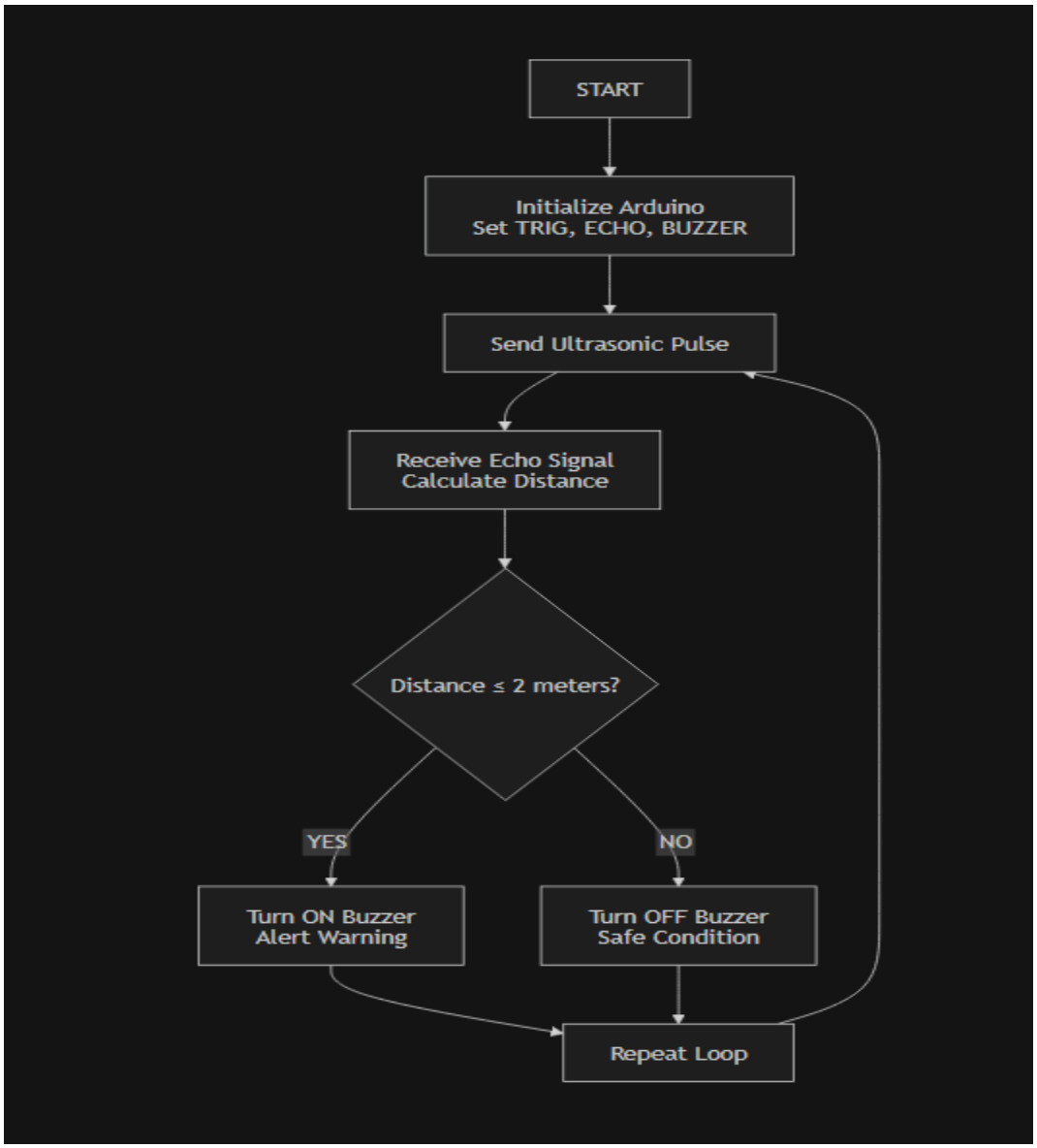
- Arduino (e.g., Arduino Uno)
- Ultrasonic Sensor (HC-SR04)
- Buzzer (alert system)
- Motor Driver Module (L298N or similar)
- DC Motor (forklift movement simulation)
- Embedded C / Arduino IDE
- Power Supply

IMPLEMENTATION:

1. The ultrasonic sensor is always sending out sound waves and then it measures how far away things are around it.
2. The Arduino takes the information from the sensor and uses it to figure out how far something is.
3. When the distance detected is more than 2 meters, the motor operates as it should.
4. If the distance is less than or equal to 2 meters:
    - The buzzer is activated
    - The motor stops immediately
5. This process repeats continuously in a loop for real-time monitoring.



Flowchart:



OUTCOMES:

This ensures that the system is able to detect any obstacles near the vehicle through its ultrasonic detector and also triggers the alarm when there is anything that is located within a distance of 2 meters from the forklift.